INCH-POUND

MIL-PRF-1/884F
7 July 1999
SUPERSEDING
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5 April 1985

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, POWER
TYPE 304TH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Triode.

See figure 1.

Mounting position: Vertical, base up or down.

Weight: 9 ounces (255.15 grams) nominal.

ABSOLUTE RATINGS: C Telegraphy.

Parameter: / Unit:	F1 MHz	Ef V	Eb V dc	Ec V dc	Ib mA dc	Ic mA dc	Pg W	Pp W
Maximum:	8	5.25 or 10.5	2,000	-300	900	140	60	300
Minimum:	---	4.75 or 9.5	---	---	---	---	---	---
Test conditions:	---	5.0 V ac	1,000	Adjust	300	---	---	---

See footnotes at end of table I.

GENERAL:

Qualification: Required.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Testing and inspection.

Inspection	Method	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
Conformance inspection, part 1		1/					
Total grid current	1266	2/		Ic	---	-25	µA dc
Peak emission	1231	---	eb = ec = 2,500 v	is	10.0	---	a
Electrode voltage (1) (grid)	1261	---		Ec(1)	-8.0	-18.0	V dc
Electrode voltage (2) (grid)	1261	---	Eb = 14 kV dc; Ib = 1 mA dc	Ec(2)	---	-1,000	V dc
Primary grid emission	1266	---	Ic = 325 mA dc; t = 15 seconds; Ef = 5.5 V ac; anode floating	Isg	---	-500	µA dc
Filament current	1301	---	Connect in parallel	If	24.0	28.0	A ac
Conformance inspection, part 2							
Power oscillation	1236	---	Eb = 3,000 V dc; Ec/Ib = 300 mA dc; F = 8 MHz; Rg = 10,000 ohms	Po	540	---	W (useful)
Amplification factor	1316	---		Mu	19.0	24.0	---
Direct-interelectrode capacitance	1331	---		Cgp	8	11	pF
				Cin	12	16	pF
				Cout	---	1	pF

See footnotes at end of table.

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TABLE I. Testing and inspection - Continued.

Inspection	Method	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
Conformance inspection, part 3							
Low-frequency vibration	1031	3/	No voltage applied	---	---	---	---
Bump	1036	3/	Angle = 10°	---	---	---	---
Life test	---	---	Group C; power oscillation; t = 500 hours	---	---	---	---
Life-test end points:	---	---					
Peak emission	1231	---	eb = ec = 2,500 v	is	8.0	---	a
Primary grid emission	1266	---	Ic = 325 mA dc; t = 15 seconds; Ef = 5.5 V ac; anode floating	Isg	---	-1.0	mA dc

1/ The acceptance level for each test listed under conformance inspection, part 1, shall be 0.65, inspection level II.

2/ This test shall be the first test performed at the conclusion of the holding period.

3/ This test shall be performed during the initial production and once each succeeding 12-calendar month period in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes having an acceptance number of zero, and a second sample of three tubes having a combined acceptance number of one. In the event of failure, the test will be made as a part of conformance inspection, part 2, with an acceptance level of 6.5, code level D. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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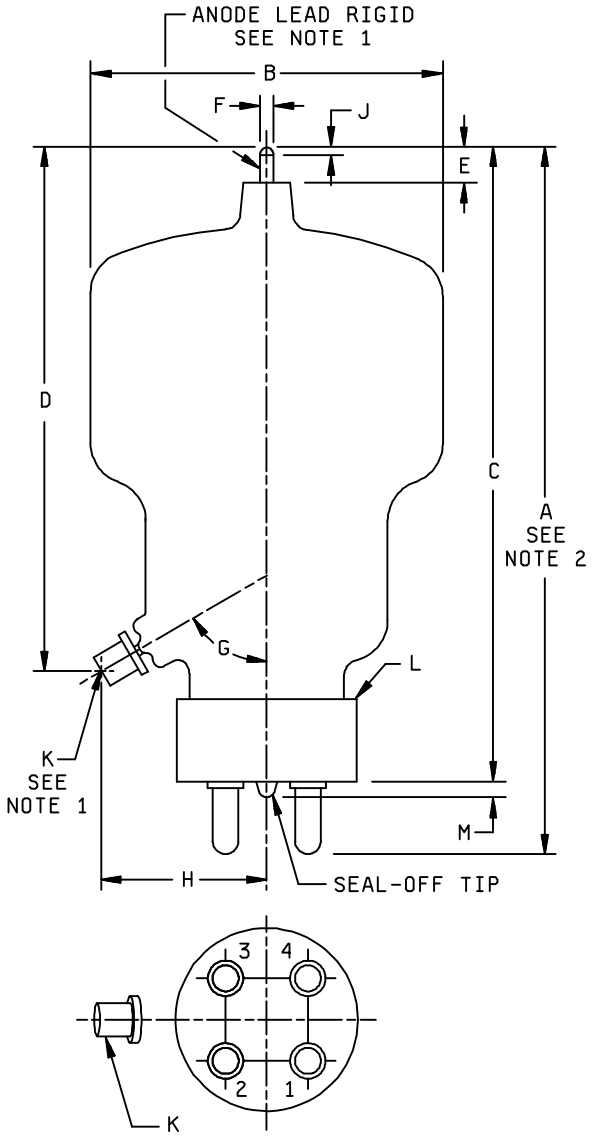

Pin connections	
Pin No.	Element
1	f1
2	f1
3	f2
4	f2
top lead	a
cap	g

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	7.250	7.625	184.15	193.68
B	---	3.563	---	90.50
C	6.500	6.875	165.10	174.63
D	5.313	5.688	134.95	144.48
E	.438	---	11.13	---
H	1.688	1.813	42.88	46.05
Conformance inspection, part 3 (see note 3)				
F	.120	.126	3.05	3.20
K	Cap: C1-23 (EIA)			
L	Base: A4-62 (EIA)			
M	---	.187	---	4.75
Reference dimensions				
G	60°		60°	
J	.031 x 45° or .005R		0.79 x 45° or 0.13 R	

NOTES:

1. Heat radiating connectors HR-6 (grid) and HR-7 (anode) or equivalent, are recommended for all applications.
2. Dimension A includes .060 inch (1.52 mm) solder tolerance on length of pins.
3. These dimensions shall be checked annually with the following sampling plan:

n1 = 4	c1 = 0	where c2 represents the total allowable failures
n2 = 4	c2 = 1	for the first and second samples combined.

In case of failure after double sampling, the failing dimension(s) shall become conformance inspection, part 2, for three consecutive successful submissions, at which time the test may revert to the conformance inspection, part 3 basis.

FIGURE 1. Outline drawing of electron tube type 304TH.

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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR
Navy - AS, CG, MC, OS
Air Force - 17, 99

Preparing activity:
DLA - CC

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